Type MS · Describe how blood clots and explain why clotting matters

Answer 1 of 10 · Question 3 part b.i · 5 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

3 (a)	<table><tr><th>Component</th><th>Name</th><th>Function</th></tr><tr><td>A</td><td>*red blood cell/erythrocyte*</td><td>*carry/transport oxygen*</td></tr><tr><td>B</td><td>white blood cell/phagocyte;</td><td>engulf bacteria;</td></tr><tr><td>C</td><td>platelets;</td><td>clotting;</td></tr></table>		4 marks
(b) (i)	any five of • enzyme/thromboplastin/thrombokinase; • released from (damaged) platelets; • prothombin to thrombin; • fibrinogen to fibrin; • meshwork formed; • trapped rbc form clot; • ref. to Ca^{2+};	reject secrete	5 marks
(ii)	• prevents loss of too much blood; • prevents entry of pathogens/bacteria; • which would cause infection/prevents infection;		3 marks

Total 12 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 1 of 10 · Question 3 part b.i · 5 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 2 of 10 · Question 9 part a.i · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

9 (a) (i)	• reference to platelets; • fibrinogen converted to fibrin; • (fibrin) mesh/crosslinks/fibres formed;		3
(ii)	• reduced blood flow to heart/cardiac muscle / tissue/cells; • less oxygen/glucose delivered; • less (aerobic) respiration/energy released (by cardiac tissue);		3
(b)(i)	size of blood clot/volume of water /aspirin solution used/temperature/shape of blood clot/width of straw;	Allow valid alternatives	1
(ii)	use water without aspirin dissolved;		1
(iii)	• the more tablets dissolved the more solution collected; • the more tablets dissolved the quicker the solution is collected; • aspirin dissolves/acts on blood clot; • size of blood clot reduced more quickly with more tablets/blood clot dissolves/breaks down faster with more tablets;		Max 3
(c)	Any four from: • aspirin binds to enzyme/aspirin competes (with substrate) for active site; • active site of enzyme changed/blocked; • less/no substrate binds (to enzyme); • reaction involving release of chemicals reduced/stopped; • platelets do not stick together;		Max 4

Total question 9 = 15 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 2 of 10 · Question 9 part a.i · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

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Type MS · Describe how blood clots and explain why clotting matters

Answer 3 of 10 · Question 5 part d · 4 marks

2020 JAN · Paper 02 · find part d in this mark scheme extract

Number				
5	(a)	• (layer 1) epidermis(1) • (layer 2) dermis (1) • (layer 3) fat/subcutaneous fat/fat layer/fat tissue/adipose layer/adipose tissue		3
	(b)	**The only correct answer is A** blood vessels and sebaceous glands ***B** is incorrect because these are in the third layer* ***C** is incorrect because these are in the third layer* ***D** is incorrect because the fat tissue is the third layer*		1
	(c)	• skin/epidermis/tissues/blood vessels damaged/open to air(1) • bacteria can enter bloodstream/body(1)		2
	(d)	Any four from: • platelets accumulate at wound/damaged area(1) • enzyme/thromokinase/thromboplastin released (1) • prothrombin converted to thrombin(1) • fibrinogen to fibrin(1) • mesh traps red blood cells(1) • scab/clot formed(1) • prevents entry of bacteria / microorganisms(1)		4

Total for Question 5 = 10 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 4 of 10 · Question 6 part a.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

6	(a) (i)	• prevent excess blood loss; • prevent entry of pathogens/bacteria/microorganisms;		2
	(ii)	V = platelets; W = thrombokinase enzyme; Y = fibrinogen; Z = fibrin;		4
	(b) (i)	• carried on X chromosome; • male must be X^hY; • female must be X^HX^h; • as allele is recessive; • but appears in next generation; any four from		5
	(ii)	• female has X^hX^h genotype; • so male offspring must be X^hY; • probability of a boy is 0.5; • probability of haemophilia is 1.0; • so probability of a boy with haemophilia is 1.0 × 0.5 = 0.5		4

Total 15 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 4 of 10 · Question 6 part a.ii · 4 marks

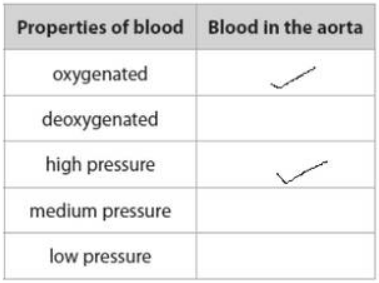
2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 5 of 10 · Question 2 part c.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number					
2	(a)	(i)	left ventricle;		1
		(ii)	Two from: • valve; • prevents the backflow of blood; • so the blood flows the correct way around the heart/in one direction;	Ignore names of valves	2
		(iii)	 Properties of blood / Blood in the aorta oxygenated ✓ deoxygenated high pressure ✓ medium pressure low pressure	One mark removed for every additional incorrect tick.	2
		(iv)	the lungs/pulmonary vein;		1
	(b)		Two from: • it could be diseased/damaged/fatty deposits/blocked arteries; • the tissue/blood type doesn't match; • it has been removed from person for too long; • wrong size;	accept named diseases allow risk of rejection	2
	(c)	(i)	C platelets;		1
		(ii)	• fibrinogen is not converted to fibrin/fibrin not produced; • so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed;		2

Total 11 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 5 of 10 · Question 2 part c.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 6 of 10 · Question 6 part a.i · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Question number				
6	(a)	(i)	any three from • prevents excessive loss of blood (1) • forms scab over wound (1) • prevents entry of bacteria /pathogens/microorganisms(1) • so prevents infection (1) • allows wound to heal (1)		3
		(ii)	• (allele) on X/sex chromosome (1) • passed on to offspring/inherited by children (1)		2
	(b)	(i)	D; (X^hY) A the genotype is not X^hX^h B the genotype is not X^HX^h C the genotype is not X^HY		1
		(ii)	B; (X^HX^h) A the genotype is not $X^h X^h$ C the genotype is not X^HY D the genotype is not X^hY		1
		(iii)	• offspring genotypes X^HX^h and X^hY (1) • 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1)	full marks for correct figures only.	2
		(iv)	• unusual for female to be haemophiliac (1) • both alleles have to be affected (1) • must have a haemophiliac father and mother with haemophilia/carrier (1)	any correct reference to excessive blood loss	3

Total for question = 12 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 6 of 10 · Question 6 part a.i · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 7 of 10 · Question 3 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
3	(a)	<table><tr><th>Component</th><th>Name</th><th>Function</th></tr><tr><td>A</td><td>red blood cell/erythrocyte</td><td>carry/transport oxygen</td></tr><tr><td>B</td><td>white blood cell/phagocyte;</td><td>engulf bacteria;</td></tr><tr><td>C</td><td>platelets;</td><td>clotting;</td></tr></table>		4 marks
	(b) (i)	any five of • enzyme/thromboplastin/thrombokinase; • released from (damaged) platelets; • prothombin to thrombin; • fibrinogen to fibrin; • meshwork formed; • trapped rbc form clot; • ref. to Ca^{2+};	reject secrete	5 marks
	(ii)	• prevents loss of too much blood; • prevents entry of pathogens/bacteria; • which would cause infection/prevents infection;		3 marks

Total 12 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 7 of 10 · Question 3 part b.ii · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 8 of 10 · Question 6 part a.i · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

6	(a) (i)	• prevent excess blood loss; • prevent entry of pathogens/bacteria/microorganisms;		2
	(ii)	V = platelets; W = thrombokinase enzyme; Y = fibrinogen; Z = fibrin;		4
	(b) (i)	• carried on X chromosome; • male must be X^hY; • female must be X^HX^h; • as allele is recessive; • but appears in next generation; any four from		5
	(ii)	• female has X^hX^h genotype; • so male offspring must be X^hY; • probability of a boy is 0.5; • probability of haemophilia is 1.0; • so probability of a boy with haemophilia is 1.0 × 0.5 = 0.5		4

Total 15 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 8 of 10 · Question 6 part a.i · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 9 of 10 · Question 1 part b · 4 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number			
1 (a)	<table><tr><th>Feature</th><th>red b c</th><th>phagocyte</th><th>lymphocyte</th><th>platelet</th><th>plasma</th></tr><tr><td>transports oxygen</td><td>✓;</td><td></td><td></td><td></td><td></td></tr><tr><td>contains haemoglobin</td><td>✓;</td><td></td><td></td><td></td><td></td></tr><tr><td>is a fluid</td><td></td><td></td><td></td><td></td><td>✓;</td></tr><tr><td>produces antibodies</td><td></td><td></td><td>✓;</td><td></td><td></td></tr><tr><td>engulfs pathogen</td><td></td><td>✓;</td><td></td><td></td><td></td></tr></table>	if more than required ticks no marks	5
(b)	any four from • forms a plug/mesh; • to prevent loss of (excess) blood; • prevents entry of bacteria/pathogens/microorganisms; • prevents infection/disease; • protects skin whilst healing;	R scab	4

Total 9 marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 9 of 10 · Question 1 part b · 4 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe how blood clots and explain why clotting matters

Answer 10 of 10 · Question 8 part b.ii · 3 marks

2020 JAN · Paper 01 · find part b.ii in this mark scheme extract

Question number					
8	(a)	(i)	X = urethra (1) Y = ureter (1)		2
		(ii)	Any two from: • transports oxygenated blood from the heart (1) • to the body/named organ (1) • maintains high blood pressure (1)		2
		(iii)	Any two from • blood in A/renal artery contains more oxygen/oxygenated (1) • blood in A/renal artery contains more urea (1) • blood in A/renal artery contains more ions (1) • similar concentration of glucose/B renal vein has less glucose (1) • blood in A/renal artery contains less carbon dioxide (1)	Allow reverse argument for B/renal vein	2
	(b)	(i)	Any four from: • dialysis fluid/dialysate contains a low(er) concentration/no urea/waste products/ions (1) • urea/waste/ions diffuse from blood (1) • through dialysis membrane (1) • from high(er) concentration (in the blood) to a low(er) concentration (in the dialysate)/down a concentration gradient (1) • dialysis membrane is partially-permeable(1)	ORA allow selectively permeable membrane	4
		(ii)	• to prevent blood from clotting whilst in the machine (1) • (added as it enters the machine) to prevent clots from blocking the machine (as blood flows through) (1) • (not added at the end) as blood clotting properties need to return to normal before passing back into patient (1)		3

Total for question 8 = 13 marks